NAMAA at KnowledgeGraphEval 2026: Entity-Span Voting and Constrained LLM Ensembles for Arabic Knowledge Graph Construction

Mahmoud Fathallah[1,5] **Mohamed Zaytoon**[1,5] **Fatimah Emad Eldin**[2,5]
Mona A.azim[3,5] **Omer Nacar**[4,5]

[1]Alexandria University [2]Trouve Labs [3]Ain Shams University
[4]Tuwaiq Academy [5]NAMAA Community
[1]es-mahmodfath96@alexu.edu.eg [1]mohamed.zaytoon24@alexu.edu.eg
[2]Fatimah@trouve.works [3]Monaabdelazim@cis.asu.edu.eg
[4]o.najar@tuwaiq.edu.sa

Abstract

Arabic knowledge graph construction is a challenging problem due to the linguistic complexity of Arabic and the need for robust generalization across domains. This paper presents our systems for the KnowledgeGraphEval 2026 shared task on Arabic knowledge graph construction and domain adaptation, covering two subtasks: AdaptNER for cross-domain named entity recognition and relation extraction (RE). For AdaptNER, we employ an entity-span ensemble combining two BIO sequence taggers with constrained conditional random field (CRF) decoding and a GlobalPointer span classifier. Our system achieved a micro-F1 score of 75.96%, ranking first in the official evaluation. For relation extraction, we use a majority-vote ensemble of three LoRA-fine-tuned causal language models. This approach achieved a micro-F1 score of 93.48%, ranking second. These results demonstrate the effectiveness of model ensembling and complementary prediction strategies for Arabic named-entity recognition and relation extraction.

1 Introduction

Knowledge graphs (KGs) structure entities and their semantic relations, which is beneficial for applications like information retrieval and question answering. The construction of KGs from Arabic texts still suffers from the language diversity and the scarcity of annotated data. KnowledgeGraphEval 2026 covers this with two subtasks: Named Entity Recognition (NER) and Relation Extraction (RE) (Aljabari et al., 2026).

The first subtask, **AdaptNER**, evaluates cross-domain generalization for Arabic NER. Systems are trained using *Wojood*, which contains approximately 550K tokens annotated with 21 entity types and nested entities (Jarrar et al., 2022; Aljabari et al., 2025a), and evaluated on ten unseen domains from *Konooz* (Hamad et al., 2025). This setup measures how well NER models handle domain shift, an important challenge since previous evaluations have reported substantial performance degradation on unseen Arabic domains.

The second subtask, **Relation Extraction**, requires predicting the semantic relation between predefined entity pairs or assigning *no-relation* when no valid relation exists (Aljabari et al., 2026). The task is based on $\text{Wojood}^{\text{Relations}}$, which contains approximately 15K annotated relation triples across 40 relation types (Aljabari et al., 2025b). Accurate relation extraction is essential for transforming detected entities into structured knowledge graph connections.

The two subtasks pose complementary challenges: AdaptNER requires accurate entity boundaries and cross-domain generalization, while RE requires distinguishing fine-grained and ambiguous relations. We address both using ensemble-based approaches that combine complementary models for entity and relation prediction.

Our main contributions are:

- For **AdaptNER**, we propose an entity-span ensemble that combines two BIO sequence taggers with constrained CRF decoding and a GlobalPointer span classifier. The system achieves a **micro-F1 score of 75.96%** and ranks **first** in the official evaluation.
- For **Relation Extraction**, we develop a majority-vote ensemble of three LoRA-fine-tuned causal language models. The system achieves a **micro-F1 score of 93.48%** and ranks **second** in the official evaluation.

2 Related Work

Subtask 1: Cross-Domain NER. Prior approaches address domain shift through neural adaptation layers (Lin and Lu, 2018), domain-adaptive pretraining on unlabeled domain text (Gururangan

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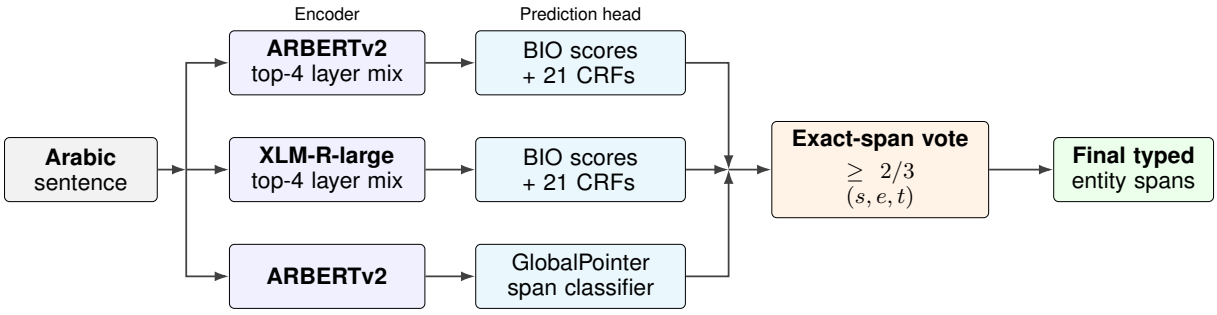


Figure 1: Subtask 1 pipeline combining two BIO-CRF models and a GlobalPointer span model using exact entity-span majority voting.

et al., 2020), and cross-domain data augmentation (Chen et al., 2021). Ensemble approaches combine predictions from multiple pretrained encoders: Schneider et al. (2022) use token-level soft voting over transformer taggers for complex NER, improving over individual models on two of the three evaluated tracks. Span-based methods provide another formulation of entity recognition; GlobalPointer directly scores typed candidate spans and supports nested entities (Su et al., 2022). For Arabic, Konooz evaluates NER across multiple domains and dialects and documents substantial performance degradation on unseen domains (Hamad et al., 2025).

Subtask 2: Relation Extraction. Entity-centered representations support supervised RE (Baldini Soares et al., 2019), while AREEj incorporates relational evidence for Arabic extraction (Al Mraikhat et al., 2024). For causal LLM-based RE, Efeoglu and Paschke (2025) fine-tune Llama2 and Mistral using LoRA and also integrate the adapted models into retrieval-augmented generation. LoRA learns low-rank weight updates while freezing pretrained parameters, reducing the number of trainable parameters (Hu et al., 2021). Heterogeneous LLM ensembling also appears in ZeroSemble, which uses weighted majority voting for entity-type resolution in a zero-shot document IE pipeline before entity-constrained RE (Pham Hoang Le et al., 2025).

3 Methodology

3.1 Subtask 1: Cross-Domain Named Entity Recognition

As shown in Figure 1, our AdaptNER system consists of three complementary models: two BIO sequence taggers with constrained CRFs and a GlobalPointer span classifier. Their predictions are combined using exact entity-span majority voting.

3.1.1 Encoder Models

We use ARBERTv2 as an Arabic-specific encoder and XLM-RoBERTa-large as a multilingual encoder (Abdul-Mageed et al., 2021; Conneau et al., 2020). To exploit information from different representation levels, we learn a weighted mixture of the top four Transformer layers:

$$h_i = \sum_{k=1}^{4} \mathrm{softmax}(\mathbf{a})_k h_i^{(k)}, \tag{1}$$

where $\mathbf{a}$ contains trainable mixture weights, h_i refers to the ith token hidden state and k refers to the layer number. The resulting representation is projected into BIO scores for each of the 21 entity types.

3.1.2 BIO-Constrained CRF

For each encoder, 21 independent linear-chain CRFs are trained, one for each entity type. Hard BIO constraints prevent invalid sequences, such as an `I` tag starting a sequence or directly following an `O` tag. Training combines the CRF negative log-likelihood with a weighted token-level cross-entropy loss:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CRF}} + 0.1\,\mathcal{L}_{\mathrm{CE}}. \tag{2}$$

Using ARBERTv2 and XLM-RoBERTa-large provides model diversity while preserving the same structured prediction architecture.

3.1.3 GlobalPointer

The third model uses ARBERTv2 with GlobalPointer (Su et al., 2022), formulating NER as typed span classification. For an entity type t and a candidate span (s, e), the span score is

$$S(t, s, e) = \frac{q_{s,t}^{\top} k_{e,t}}{\sqrt{d}}, \tag{3}$$

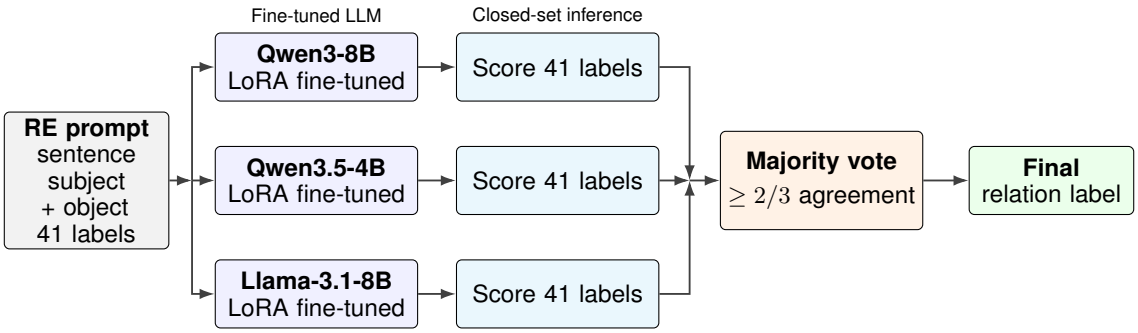


Figure 2: Subtask 2 pipeline combining three LoRA-fine-tuned causal LLMs with constrained label scoring and majority voting.

where $q_{s,t}$ and $k_{e,t}$ are learned type-specific representations of the start and end positions, respectively, and d is their representation dimension. The dot product measures the compatibility of s and e as the boundaries of an entity of type t, while the factor $\sqrt{d}$ stabilizes the score magnitude. Thus, a higher $S(t, s, e)$ indicates greater confidence that the span corresponds to entity type t.

The maximum span length is a hyperparameter set to 64 words, based on the assumption that longer named entities are rare. At inference, spans whose scores exceed the decoding threshold are retained. If more than 256 candidates remain for an entity type, only the top 256 are considered. Weighted interval scheduling is then applied to select the highest-scoring non-overlapping spans for each entity type.

3.1.4 Exact Entity-Span Ensemble

The three models are combined using exact entity-level majority voting. Each model prediction is converted into a set of typed spans of the form $(\text{start}, \text{end}, \text{entity type})$, and votes are counted for each unique span. A candidate is retained when at least two of the three models predict exactly the same start and end boundaries and entity type. This span-level voting directly matches the official exact-span evaluation criterion and avoids inconsistencies that may arise from combining predictions at the token level.

3.2 Subtask 2: Relation Extraction

As shown in Figure 2, our relation extraction system uses supervised fine-tuning followed by closed-set relation label scoring. We fine-tune Qwen3-8B, Qwen3.5-4B[1], and Llama-3.1-8B-Instruct using Low-Rank Adaptation (LoRA) (Hu et al., 2021). Each prompt contains the Arabic sentence, ordered subject and object mentions, and the complete relation-label inventory, while the target contains only the correct relation label.

3.2.1 LLM Fine-Tuning and Ensemble

Rather than using free-form generation, inference is constrained to the 41 valid relation labels. Each candidate relation r is scored using its mean teacher-forced token log-likelihood:

$$\frac{1}{|r|}\sum_{j=1}^{|r|}\log p(r_j \mid x, r_{<j}), \tag{4}$$

where r_j is the j-th token of the candidate relation label and x denotes the input prompt. Averaging over the label length reduces bias toward labels represented by fewer tokens. The highest-scoring label is selected for each model. The predictions from the three LLMs are then combined by majority voting with a deterministic tie-breaker, with Qwen3.5-4B used as the tie-breaker when all predictions differ, as it is the model with best standalone performance.

4 Experiments

4.1 Datasets

For **Subtask 1**, we train on the Wojood NER corpus, which contains approximately 550K Arabic tokens annotated with 21 entity types, including nested entities (Jarrar et al., 2022). Evaluation is performed on ten unseen domains from Konooz, making the task a direct test of cross-domain generalization (Hamad et al., 2025).

For **Subtask 2**, we use the Relation Extraction dataset derived from $\text{Wojood}^{\text{Relations}}$, which contains approximately 15K annotated relation triples across 40 semantic relation types (Aljabari et al., 2025b). Including `no_relation`, the final prediction space contains 41 labels.

[1]Official model identifier: `Qwen/Qwen3.5-4B`.

Model	Acc.	Recall	Precision	Micro-F1
Shared-task baseline	–	63.87	62.30	63.00
ARBERTv2 + CRF	99.55	70.08	80.69	75.01
XLM-R-large + CRF	99.55	71.12	79.41	75.04
ARBERTv2 + GlobalPointer	99.48	66.17	84.02	74.03
Exact-Span Ensemble	99.53	69.44	83.83	**75.96**

Table 1: Official AdaptNER test results. Precision, recall, and F1 are entity-level micro-averages under exact span-and-type matching. Scores are reported as percentages; – denotes an unreported baseline metric.

4.2 Training Setup

For AdaptNER, models use a maximum input length of 512, dropout of 0.1, batch size of 32, BF16 precision, and train for up to 15 epochs with early stopping patience of 5. The encoder learning rate is 2×10^{-5}, with 10^{-4} for task heads and 10^{-3} for CRFs; we use 0.01 weight decay, 10% warmup, and a linear scheduler.

For Relation Extraction, the three causal LLMs are fine-tuned with LoRA using rank 16, scaling factor 32, dropout 0.05, and a learning rate of 2×10^{-4}. Training uses BF16 precision for three epochs, with sentence-level grouping between training and validation splits.

4.3 Results

4.3.1 Subtask 1: AdaptNER

Table 1 compares the shared-task baseline, the two CRF systems, the GlobalPointer model, and their exact-span ensemble. The ensemble achieved a **micro-F1 score of 75.96%**, exceeding the baseline F1 of 63.00% by 13.00 percentage points and ranking **first** in the official evaluation.

4.3.2 Subtask 2: Relation Extraction

Table 2 compares the shared-task BERT (ArBERTv2) and LLM (gpt-4o) baselines with the three fine-tuned LLMs and their majority-vote ensemble. The ensemble achieved a **micro-F1 score of 93.48%**, exceeding the baseline F1 scores of 88.61% and 85.78% by 4.87 and 7.70 percentage points, respectively, and ranking **second** in the official evaluation.

4.4 Analysis & Limitations

For AdaptNER, exact-span voting enhances micro-F1 from 75.04 to 75.96, a gain of 0.96 points. Ensemble precision rises from 79.41 to 83.83, while recall drops from 71.12 to 69.44, indicating a conservative approach that may filter out valid entities. In Relation Extraction, majority voting boosts the micro-F1 score from 92.77 to 93.48, reflecting a gain of 0.71 points, along with improved precision (96.20 to 96.80) and recall (89.58 to 90.38). However, both ensemble methods increase inference costs and do not resolve shared errors, with limitations on error localization.

Model	Recall	Precision	Micro-F1
Shared-task baseline	–	–	93.09
Qwen3-8B	89.35	95.59	92.36
Qwen3.5-4B	89.58	96.20	92.77
Llama-3.1-8B-Instruct	89.03	96.21	92.48
Majority-Vote Ensemble	90.38	96.80	**93.48**

Table 2: Results for Subtask 2 (Relation Extraction). Scores are reported as percentages.

5 Conclusion

This paper presented our systems for the KnowledgeGraphEval 2026 shared task on cross-domain Arabic NER and relation extraction. For AdaptNER, an ensemble of two BIO-CRF models and a GlobalPointer classifier achieved **75.96%** micro-F1 and ranked **first**. For Relation Extraction, a majority-vote ensemble of three fine-tuned causal LLMs achieved **93.48%** micro-F1 and ranked **second**. Overall, the results highlight the effectiveness of complementary model ensembles for Arabic entity and relation extraction.

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